Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Metric-Semantic Mapping and Neuro-Symbolic Methods for Scene Reasoning in Support of Assistive Robot Task Completion			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power (), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), **Advanced Robotics & Manufacturing (X)**, Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical / Materials		Mechanical / Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics / Automation Machinery	Robot Control and Intelligence Technologies	Robotics / Automation Machinery	Other Robotics/Automation Machinery Technologies
Concept of Proposed Technology	The proposed technology consists of a hierarchical, multi-layered metric-semantic mapping system that stores robot retrievable information about a mapped environment. To interpret this information is a layered, neuro-symbolic task reasoning and planning pipeline that leverages the inductive strengths of deep neural networks, also known as connectionist systems, with the analytical strengths of symbolic planning methods, simply called symbolic systems. If desired, the system may be connected to a translator for converting the robotic maps into human interpretable and annotated maps compatible with the OpenStreetMap format and viewable from web browsers based on rendering engines that interpret the OSM format or alternative formats upon conversion, such as GeoJSon, IMDF (Apple), and KML (Google). The combined system can be used for a variety of robotic use cases, such as indoor infrastructure inspection and asset management; assistive robotics for the home or office; indoor mapping and progress monitoring; warehouse logistics support and manufacturing; and other physical tasks for which			

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	worker presence brings a level of risk that could be reduced through the use of robot substitutes. The mapping system and reasoning framework builds on a decade of research into high performance robot autonomy covering indoor navigation with strong collision-free guarantees; robust, real-time indoor SLAM systems and indoor mapping; learning-enabled control theory and real-time safety constraint synthesis from perceptual data; and vision-based manipulation. Binding all of these modules together is a perception-oriented, metric-semantic mapping system that stores sensory data in a multi-layered representation that encodes the data needed for navigation mapping, localization, manipulation, and scene reasoning. Reproducing the above paragraph in bullet form, the main components of the system include: ☐ Real-time, indoor navigation system with strong collision-free guarantees and 3D collision checking; ☐ Real-time, indoor SLAM (LIDAR or passive stereo) with robust multi-sensor fusion; ☐ High performance grasp manipulation processing based on deep networks and augmented by manipulation motion planners for improved collision safety; ☐ Real-time synthesis of safety constraints for navigation from perceptual data, with possibility to extend to manipulation; ☐ Perception-space data representation and processing schemes for real-time collision recognition and safety intervention; ☐ Neuro-symbolic reasoning pipeline that interleaves connectionist and symbolic systems for hallucination rejection, certified falsification, and high task planning accuracy; and ☐ An internal metric-semantic mapping system for storing data pertinent to all of the above robot autonomy components. The above features all support mobile robot and mobile-manipulation robot needs. Partnerships may involve all or a subset of these features as needed by the application domain and embodied structure of the best fit robot solution.
Rationale for Joint R&D Project	Collaboration is required to translate the current research outcomes to commercializable robotics application areas with a company with more expertise in the given market. A company who has the domain knowledge to direct the underlying implementation and maturation of methods towards market needs, and to develop a more appealing form factor and feature set compatible with customer expectations. The company also has more experience creating surrogate environments that can best stress test robotic solutions prior to deployment. Importantly, the company understands the economics of the product and what richness of technology integration the market would support in relation to final build costs. The deployment of robots into complex and evolving environments is non-trivial. Current

	solutions that do so often bring more work, known as robot wrangling or robot tending. By working together, GT and the company, can work towards robust solution that fist customer needs in a cost-effective manner, while mitigating the pitfalls of modern robot deployments. Though many of the elements describing the concept have been shown to function well in laboratory settings, additional insights and deployment experience is needed to arrive at real-world feasible solutions. Partnering with a company who has strong insights into deployment needs and specifications that can help guide the GT team's efforts. Furthermore, they can provide the domain expertise needed to fully realize the autonomy software stack of the robot for the given application area. If human interpretable maps and user interfaces are required, the company would provide the expertise and UI design capabilities needed to realize the user interface. The GT team would coordinate with the company to ensure that the data provided meets the database requirements of the browser-based or app-based interface designed by the company. Conversely, the GT team has extensive experience across the breadth of robot autonomy, as related to perception-informed planning, control, and decision-making in the area of robotics. The GT team can provide important insights to the company as they work towards a minimally viable product or an improved product iteration. The PI has worked on projects aimed at higher TRL levels and understands how to coordinate with companies to enable this upwards progress.
Key Words	1. Assistive and service robotics 2. Metric-topological and semantic mapping 3. Neuro-symbolic reasoning 5. Robot scene understanding 6. Robot autonomy

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]		**Start of the Project**	Level (2-4)	**End of the Project**	Level (5-6)
R&D Objectives		The R&D objectives will depend on the company partner. Candidate partners are expected to be interested in developing an indoor mobile robot or a mobile-manipulation robot that will complete specific tasks within the indoor environment. Outdoor environments are possible to include, but would require additional research activities to realize. Given this general conception, the main objectives are to ☐ Specify and develop an indoor metric-semantic mapping solution compatible with the target robot application;			

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		☐ Specify and develop scene and task reasoning pipelines based on the metric-semantic map; ☐ Create customized robot software (navigation, mapping, localization, manipulation) building on the GT team's capabilities and integrating with the company's capabilities that will be highly performant in the target environments (e.g., high levels of robustness to environment and task completion); ☐ Develop perception-informed safety reasoning and modeling algorithms compatible with the robot task (navigation or manipulation); ☐ Improve the underlying learning-enabled components and symbolic planning methods targeted to the application domain; and/or ☐ Perform system integration to realize the defined product and run test deployments in suitable environments. Emphasis or priority of the above will be determined by the partnering company.
R&D Target Specifications		Depending on the target application, all or a subset of the following will be demonstrated: ☐ Collision-free navigation in the target environments (0% collision rate); ☐ Repeatable navigation to target locations within the environment (100% success at 40cm error; 80% success at 20cm error) ☐ High performance grasping and manipulation (90% or higher task completion rate at first attempt) with high falsification detection (95% or higher) for correcting failed grasping attempts. Ideally, above 95% for manipulation success rates. ☐ High performance task reasoning (100% falsification rate for incorrectly specified tasks, 90% or higher completion rates for correctly specified tasks) ☐ Sequential task reasoning and completion rates above 95% for simple tasks, above 90% for non-trivial tasks, and above 80% for sequential tasks. ☐ Above 90% success rate at object acquisition, localization, and navigation tasks. ☐ The primary limitation preventing perfect completion will be the object recognition method, as measured from test trials. Demonstrated by a low ratio of (failure due to system)/(failures due to object recognition), where low is defined as less than 1/10. ☐ Demonstration that online safety modeling captures the safe region by rejecting all of the unsafe regions when the models are used for motion planning.
R&D Plan		The research approach to take will depend on the partnering company. The GT Team anticipates that the research approach will based on a neuro-symbolic approach, whereby connectionist system elements and symbolic system elements are combined and interleaved to arrive at a high performance system with strong performance guarantees regarding decision making. The metric-semantic mapping and environment modeling framework will be established in partnership with the company and a research approach defined based on the application needs. The key tasks will be derived from the established research approach. As part of the partnership, the anticipated progress is as follows: **Year 1 Milestones:** - Define problem of interest and fully specify robot features and tasks.

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		- Specify the metric-semantic map or compatible internal robot environment representation as suited to the task, and demonstrate proof-of-concept implementation of the map / representation. - Specify the robot autonomy stack and demonstrate a minimum viable implementation of the full stack. - Evaluate performance of preliminary implementation and establish incremental performance goals that will define successful progress. - Identify candidate deployment environments and test robot in them to generate baseline performance measurements and outcomes. **Year 2 Milestones:** - Refinement of algorithms based on deployment environment and baseline performance outcomes. - Complete specification of tasks and setup of realistic test environments along with means to reliable evaluate performance outcomes. - Coordination of company and GT to integrate and improve the baseline autonomy stack. - Specify and test out human user interfaces for deploying robot as needed for target application tasks. **Year 3 Milestones:** - Further improvement of baseline methods and advances based on established research methodology from Year 1 specifications. - Finalizing of TRL Level 5-6 product and testing in appropriate environments. - Identify critical barrier to commercialization and target improvements to remedy the barriers. - Candidate test site identified and trial runs implemented.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.*
	
	The GT team contributions will include the following.
	- Baseline implementation and iterative development of robot autonomy stack as needed, inclusive of:
	  - Collision-free navigation;
	  - Passive stereo or LIDAR-based mapping and localization algorithms;
	  - Manipulation scene understanding, motion planning, and collision-free safety module;
	  - Creation of metric-topological environment map;
	  - Development and refinement of neuro-symbolic approach to task reasoning and planning;
	  - Scene translation to human interpretable formats as needed;
	- Baseline implementation on laboratory robots equipped with target sensory configuration and specified robot actuation designs.
	- Development of task-relevant performance metrics in target deployment environments, and compatible validation methodology.
	- Research coordination with company.
	- Technical guidance and collaboration with industrial partner.

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Expected R&R of Korean SME Partner	The industrial partner, a Korean SME, will contribute to the project as per below; - Translate preliminary design to a more commercial design meeting customer needs; - Fabrication commercial robot design and create functional prototypes; - Assist with systems integration and iterative development of the target robot autonomy stack; - Provide own custom algorithms specialized to targeted task, especially any involving object recognition or image-based recognition as needed for task execution and completion; - Provide domain insights and ideate with GT team to establish effective software design specifications for a successful product; - Identify of and provide access to TRL 5-6 testing environments; - Test and validate the proposed product; - Confirm that design meets specified success criteria.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The primary achievement will be demonstration at TRL Level 5 or 6 of the intended robotic product, along with a post-analysis of critical features to address and a report of the commercialization potential of the prototype. In terms of the fundamental technology, the expected outcomes are to: - Validate the effectiveness of neuro-symbolic reasoning pipelines within robot autonomy stacks in terms of robustness to variable input, rejection of invalid input, and high success rate for valid inputs; - Verify the utility of the derived metric-topological mapping scheme in relation to the target tasks; - As fitting, confirm that the translated human interpretable map information suits the needs of human users and has high usability; - Identify a commercialization strategy covering the next 3-5 of development and TRL improvement. The project will provide the industry partner with a high performance AI-enabled and symbolically validated robot autonomy stack whose underlying perception processing pipeline enable real-time operation on commercially available hardware. The partnership will strengthen the commercial competitiveness of Korean robot-using or robot-provisioning SMEs.

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☐ **Terms:**

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization